

8.2 SPAN TABLES

These span tables are based on typical residential construction using No. 2 grade lumber and standard loading conditions. **Always verify with your local building department and engineer's specifications for your specific project.** Spans are in feet-inches.

FLOOR JOISTS - 40 PSF Live Load, 10 PSF Dead Load

No. 2 Grade Lumber, Deflection Limited to L/360

Joist Size: 2x8	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	13'-3"	12'-0"	11'-2"	9'-11"
Douglas Fir-Larch	12'-8"	11'-6"	10'-8"	9'-6"
Hem-Fir	12'-2"	11'-0"	10'-3"	9'-1"
Spruce-Pine-Fir	11'-10"	10'-9"	10'-0"	8'-11"

Joist Size: 2x10	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	16'-11"	15'-4"	14'-3"	12'-8"
Douglas Fir-Larch	16'-2"	14'-8"	13'-7"	12'-1"
Hem-Fir	15'-6"	14'-1"	13'-1"	11'-7"
Spruce-Pine-Fir	15'-1"	13'-8"	12'-9"	11'-4"

Joist Size: 2x12	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	20'-7"	18'-8"	17'-4"	15'-5"
Douglas Fir-Larch	19'-8"	17'-10"	16'-7"	14'-9"
Hem-Fir	18'-10"	17'-1"	15'-11"	14'-1"
Spruce-Pine-Fir	18'-4"	16'-8"	15'-6"	13'-9"

Notes:

- O.C. = On Center (spacing between joist centers)
- Spans assume single span with bearing at ends only
- For cantilevers, maximum overhang = 1/4 of back span
- Add blocking or bridging at mid-span for joists over 8 feet

CEILING JOISTS - 10 PSF Live Load, 5 PSF Dead Load

No. 2 Grade Lumber, Not Supporting Habitable Space

Joist Size: 2x6	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	17'-4"	15'-9"	14'-8"	13'-0"
Douglas Fir-Larch	16'-7"	15'-1"	14'-0"	12'-5"
Hem-Fir	15'-11"	14'-5"	13'-5"	11'-11"
Spruce-Pine-Fir	15'-6"	14'-1"	13'-1"	11'-8"

Joist Size: 2x8	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	22'-11"	20'-9"	19'-4"	17'-2"
Douglas Fir-Larch	21'-11"	19'-11"	18'-6"	16'-5"
Hem-Fir	21'-0"	19'-1"	17'-9"	15'-9"
Spruce-Pine-Fir	20'-6"	18'-7"	17'-4"	15'-5"

Joist Size: 2x10	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	Max Span	26'-6"	24'-8"	21'-11"
Douglas Fir-Larch	Max Span	25'-5"	23'-8"	21'-0"
Hem-Fir	26'-10"	24'-4"	22'-8"	20'-2"
Spruce-Pine-Fir	26'-2"	23'-9"	22'-1"	19'-8"

COMMON RAFTERS - 30 PSF Live Load, 10 PSF Dead Load

No. 2 Grade Lumber, Snow Load Areas

Rafter Size: 2x6	12" O.C.	16" O.C.	24" O.C.
Southern Pine	13'-7"	12'-4"	10'-11"
Douglas Fir-Larch	13'-0"	11'-10"	10'-6"
Hem-Fir	12'-6"	11'-4"	10'-1"
Spruce-Pine-Fir	12'-2"	11'-1"	9'-10"

Rafter Size: 2x8	12" O.C.	16" O.C.	24" O.C.
Southern Pine	17'-11"	16'-3"	14'-5"

Douglas Fir-Larch	17'-2"	15'-7"	13'-10"
Hem-Fir	16'-6"	14'-11"	13'-3"
Spruce-Pine-Fir	16'-0"	14'-7"	12'-11"

Rafter Size: 2x10	12" O.C.	16" O.C.	24" O.C.
Southern Pine	22'-10"	20'-9"	18'-5"
Douglas Fir-Larch	21'-11"	19'-11"	17'-8"
Hem-Fir	21'-0"	19'-1"	16'-11"
Spruce-Pine-Fir	20'-6"	18'-7"	16'-6"

HEADERS - Load Bearing Walls

Building Width 24 feet, One Story Above

Opening Width	Single Story	Two Stories	Three Stories
Up to 3 feet	2x6	2x8	2x10
4 feet	2x8	2x10	2x12
5 feet	2x10	2x12	2-2x12
6 feet	2x12	2-2x10	2-2x12
8 feet	2-2x10	2-2x12	3-2x12
10 feet	2-2x12	3-2x12	4-2x12
12 feet	3-2x12	4-2x12	Engineered Beam

Notes:

- Headers must be installed on edge with crown up
- Use solid blocking or cripple studs below header
- King studs and jack studs required on both sides
- For wider openings or different conditions, consult engineer

DECK JOISTS - 40 PSF Live Load, 10 PSF Dead Load

Southern Pine or Douglas Fir-Larch No. 2 Grade

Joist Size: 2x6	12" O.C.	16" O.C.	24" O.C.
Maximum Span	9'-9"	8'-10"	7'-10"

Joist Size: 2x8	12" O.C.	16" O.C.	24" O.C.
Maximum Span	12'-10"	11'-8"	10'-4"

Joist Size: 2x10	12" O.C.	16" O.C.	24" O.C.
Maximum Span	16'-5"	14'-11"	13'-3"

Joist Size: 2x12	12" O.C.	16" O.C.	24" O.C.
Maximum Span	20'-0"	18'-2"	16'-1"

Deck Construction Notes:

- Joist spans measured from face of ledger to centerline of beam
- Use pressure-treated or naturally decay-resistant lumber
- Install joist hangers at ledger connections

- Maximum cantilever = 1/4 of joist span
- Check local codes for additional requirements

IMPORTANT DISCLAIMER: These tables are for general reference only and represent common residential construction conditions. Actual span requirements may vary based on lumber species, grade, moisture content, load duration, and local code requirements. Always consult with your building department and structural engineer for your specific application. Engineered lumber products (I-joists, LVL, etc.) have different span capabilities - refer to manufacturer's specifications.